

REMORA

A Reference Architecture for Governed Agentic AI

Evidence-based whitepaper for runtime AI assurance, selective trust, policy-gated action, and auditable agent governance.

Prepared for
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Evidence baseline
Uploaded technical report + public REMORA
repo head 5e051b9 (v0.6.0)

Status
Technical whitepaper - evidence
scoped, not production
certification

Table of contents

Table of contents	2
1. Executive thesis	4
Strategic conclusion	5
2. Evidence scope and standard of proof	5
Evidence classes	5
3. Problem: agentic AI turns answers into actions	6
Minimum control-plane requirements	6
4. Contributions	6
5. System model	6
Decision outputs	7
6. Mathematical and decision-theoretic model	7
6.1 Entropy and dissensus	7
6.2 Temperature-like uncertainty proxy	7
6.3 Free-energy proxy	7
6.4 Lyapunov-style session monitor	7
7. Architecture and runtime components	8
7.1 Cascade execution path	8
7.2 Runtime capability inventory	9
8. Enterprise governance primitives	9
8.1 Policy-as-code	9
8.2 Audit graph and observability	10
8.3 Tool-call safety	10
8.4 Alignment with governance frameworks	10
9. Empirical evaluation	10
9.1 Full-coverage QA is not the headline	10
9.2 Selective trust is the QA win	11
9.3 Tool-call safety is the commercial center of gravity	11

9.4 Calibration and negative results	12
10. v0.6.0: quality-lift analysis	12
10.1 PlattScaler	12
10.2 OracleDiversityTracker	12
10.3 DomainCoverageOptimizer	13
10.4 Epistemic / aleatoric uncertainty decomposition	13
10.5 MoA Synth	13
11. Claim ledger and public claim boundary	13
12. Threat model for governed agents	13
Production control requirements	14
13. Product interpretation and market category	14
Initial wedges	14
14. Roadmap to world-class credibility	14
14.1 Technical priorities	15
14.2 External validation protocol	15
15. Final conclusion	15
Appendix A. Source notes and repository evidence map	16
Appendix B. External references	17
Appendix C. Minimal production decision packet	17
Appendix D. Suggested release gate	17

1. Executive thesis

Core thesis

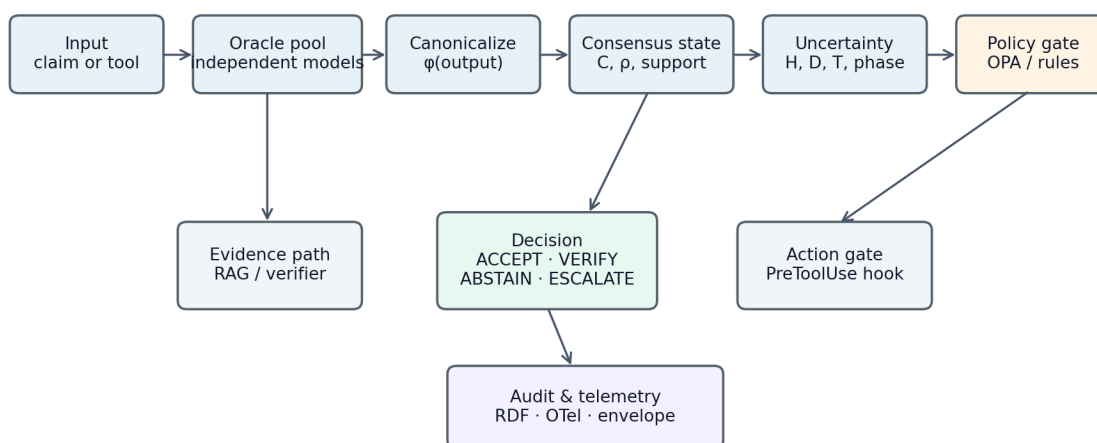
REMORA should be positioned as a governance control plane for agentic AI: it decides when probabilistic model output is reliable enough to answer, when evidence must be consulted, when action must be blocked, and when a human must approve. Its strongest validated value is selective reliability and safe tool execution - not universal full-coverage QA superiority.

The uploaded technical report already reached the correct strategic conclusion: REMORA is not merely a multi-model voting wrapper; it is a conservative AI assurance layer that treats model output as an unsafe proposal until trust, evidence and policy checks permit release or action. [S1] This whitepaper rewrites that report into a publication-grade structure, updates it with the latest inspected v0.6.0 repository evidence, and separates supported claims from candidate theory.

The system is most compelling where the cost of a false positive is higher than the cost of escalation: infrastructure change control, legal/document analysis, compliance evidence generation, incident response, MCP tool safety, and AI-assisted operations. In those environments, the right metric is not answer volume; it is governed action quality.

REMORA as a governed agentic AI control plane

Model output is treated as a proposal until trust, evidence and policy permit action.



Operational thesis: make the decision boundary explicit, measurable and auditable - not merely more verbose.

Figure 1. REMORA control-plane architecture.

Source note: synthesized from REMORA cascade, policy, audit and tool-hook modules; see notes [S3]-[S12].

Strategic conclusion

Question	Evidence-scoped answer
What is REMORA?	A runtime assurance layer for agentic AI that converts model proposals into governed decisions: ACCEPT, VERIFY, ABSTAIN or ESCALATE.
What is the strongest claim?	REMORA can identify high-precision answer slices at reduced coverage and can block unsafe tool execution in its benchmark-scoped policy-gate tests.
What should not be claimed yet?	Production-proven safety, universal hallucination bounds, full-coverage superiority over majority vote, or a complete physical law of LLM consensus.
What changed in v0.6.0?	New calibration, oracle diversity, domain-threshold, uncertainty decomposition and MoA synthesis components were added; the next step is runtime coupling and external validation.

2. Evidence scope and standard of proof

This document uses a conservative evidence standard. Repository claims are treated as supported only where a code path, committed artifact or test path exists. Commit messages are useful for audit chronology, but they are not independent proof. External production claims require external datasets, fresh model calls, frozen thresholds and independently reproducible runs.

The original report is based on a static repository and artifact review, with an attempted pytest run that timed out before all slow replay-style tests completed. The report explicitly treats committed result artifacts and repository tests as project evidence rather than independent reproduction. [S1] That limitation is preserved here.

Evidence classes

Evidence class	Accepted meaning in this whitepaper	Example
Implemented	A concrete source module exists and exposes an API.	PlattScaler, OracleDiversityTracker, OPAAAdapter.
Tested	A test file or test reference exists for the component.	test_platt_scaler.py, test_moa_synthesizer.py.
Artifact-backed	A committed result JSON, report or claim ledger entry supports the number.	selective trust, tool-call v2 results.
Benchmark-scoped	The result is valid for the included benchmark design, not a universal guarantee.	0.0% unsafe execution in v2 simulator.
Candidate	A theoretical or empirical idea with implementation but insufficient validation.	hallucination-bound proxy, strong thermodynamic-law claims.

Audit warning

The latest repository head reports 1,184 tests passing in the v0.6.0 commit message. This whitepaper records that as a repository-reported status, not as an independently executed result in this environment. [S2]

3. Problem: agentic AI turns answers into actions

Classical chatbot quality problems are mostly epistemic: a model may answer incorrectly. Agentic systems add operational risk: a wrong answer may become a shell command, API mutation, network change, payment instruction, legal assertion or production deployment. A governance layer must therefore decide not only what the model said, but whether the present evidence and policy state allows the system to act.

This is the governance gap REMORA addresses. It inserts an explicit decision boundary between probabilistic reasoning and deterministic execution. That boundary must be measurable, auditable, policy-driven and fail-safe.

Minimum control-plane requirements

- 1 Selective acceptance: return answers only when the trust state justifies action or response.
- 2 Evidence escalation: move from parametric model memory to external evidence when consensus is insufficient.
- 3 Policy precedence: hard safety rules override model confidence and voting outcomes.
- 4 Action isolation: tool calls are proposals; execution requires a separate allow decision.
- 5 Auditability: every route and refusal must be replayable from decision metadata.

4. Contributions

REMORA is best read as a system contribution rather than a single algorithm. It combines known families of methods - multi-agent consensus, selective prediction, confidence calibration, policy-as-code, evidence retrieval and tool-use gating - into one governance architecture.

Contribution	Description	Evidence level
Consensus as state	Model agreement is measured over canonical claims, not raw text or a bare vote.	Implemented core engine
Thermodynamic signals	Entropy, dissensus, temperature-like proxy and phase labels are used as routing signals.	Implemented; heuristic interpretation
Conservative routing	The system has first-class VERIFY, ABSTAIN and ESCALATE outputs.	Implemented and benchmarked
Policy-gated action	Tool calls are classified and may be blocked before execution.	Implemented and benchmarked
Enterprise governance primitives	OPA, RDF audit, OTel observability, claim ledger and evidence artifacts.	Implemented/prototype
v0.6.0 quality lift	Calibration, diversity, domain thresholds, uncertainty decomposition and MoA component.	Implemented + tested components; runtime wiring varies

5. System model

Let a request q produce model outputs $y_1 \dots y_n$. REMORA first maps each output to a canonical claim or verdict state $c_i = \phi(y_i)$. This is essential: exact string agreement is too brittle, while unbounded semantic interpretation is too hard to audit. The canonical state becomes the measured object for consensus, policy and audit.

The decision function can be expressed as:

D = g(C, U, P, E, R)

where C is canonical consensus state, U is uncertainty state, P is policy observation, E is evidence status and R is operational risk. The output D is one of ACCEPT, VERIFY, ABSTAIN or ESCALATE.

Decision outputs

Output	Meaning	Operational posture
ACCEPT	The answer or action passes trust, evidence and policy conditions.	Proceed and audit.
VERIFY	The case is answerable but requires evidence, review or additional checks.	Do not treat as final.
ABSTAIN	The system withholds because the risk of false confidence is too high.	Return uncertainty or refuse.
ESCALATE	The case requires human or domain-expert approval.	Route to operator workflow.

6. Mathematical and decision-theoretic model

The mathematics should be presented as operational signal engineering, not as a claim that LLMs literally form a physical thermodynamic system. The phase vocabulary is useful because it compresses agreement structure into routing signals. It should remain explicitly evidence-scoped.

6.1 Entropy and dissensus

For a set of canonical answer states with empirical probabilities p_j , Shannon entropy is:

$$H = - \sum_j p_j \log_2(p_j)$$

A simple dissensus measure is:

$$D = 1 - \max_j p_j$$

H captures distributional spread; D captures the complement of the strongest support. Both are bounded and interpretable for routing, but neither alone is a truth guarantee.

6.2 Temperature-like uncertainty proxy

REMORA uses an effective temperature-like quantity T as an engineering proxy for uncertainty or disorder. It is assembled from observable quantities such as entropy, support, confidence dispersion and related signals. The safe description is: T is a calibrated routing feature, not a physical temperature.

6.3 Free-energy proxy

The repository uses a Helmholtz-style proxy:

$$F_{\text{proxy}}(T) = \lambda D - T H$$

The expression is valuable as a compact way to combine disagreement cost with entropy. It should be described as a free-energy-inspired control objective unless a proof is stated with explicit assumptions and a clear mapping from oracle distributions to a Gibbs or maximum-entropy model. Stronger repository notes now explore MaxEnt grounding, but public claims should remain careful until externally reviewed.

6.4 Lyapunov-style session monitor

For session stability, REMORA tracks:

$$V(t) = H(t) + \lambda D(t)$$

A non-increasing V(t) trajectory is a monitor for whether entropy and dissensus are not growing across a session. It is not a proof of general agent correctness. The v0.6.0 governance framing should preserve this distinction.

Mathematical guardrail

The mathematical language is strongest when tied to observed routing behavior: low-temperature / ordered regimes are higher precision in the committed benchmarks; disordered states justify abstention or escalation. It is weakest when framed as a universal law of LLM cognition.

7. Architecture and runtime components

The architecture is a nested governance layer around model outputs and tool calls. It is not intended to replace foundation models; it constrains how their outputs become decisions.

Governance stack: from probabilistic proposals to auditable action

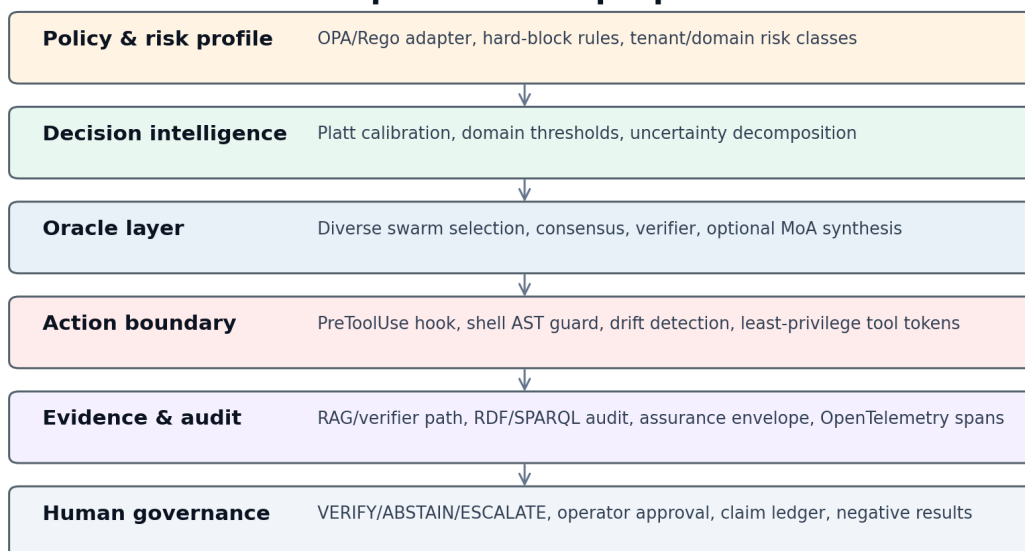


Figure 2. Governance stack from policy to audit.

Source note: synthesized from latest inspected REMORA modules and governance design; see notes [S5]-[S12].

7.1 Cascade execution path

The inspected CascadeEngine initializes and executes FastGate, ConsensusGate, VerifierGate and SelfConsistencyGate, with CritiqueRevisionGate invoked conditionally after verifier challenge. It supports oracle-call budgets and safe VERIFY return when the budget is exhausted. [S3]

The latest commit introduces a MixtureOfAgentsSynth component and a CascadeStage.MOA_SYNTH enum. The component is implemented and tested, but in the inspected engine it is not yet wired into the default stage_runners list. For accuracy, this whitepaper treats MoA as an optional implemented component rather than a default six-stage runtime path. [S4]

7.2 Runtime capability inventory

Capability	Repository component	Whitepaper status
Fast confidence gate	FastGate	Integrated
Multi-oracle consensus	ConsensusGate + Remora engine	Integrated
Independent verifier	VerifierGate + LLMJudge	Integrated
Critique/revision	CritiqueRevisionGate	Conditionally integrated
Self consistency	SelfConsistencyGate	Integrated terminal stage
MoA synthesis	MixtureOfAgentsSynth	Implemented/tested component; not default engine path
Policy-as-code	OPAAdapter	Implemented adapter
RDF audit export	RDFAuditExporter	Implemented exporter
OpenTelemetry	RemoraTracer	Optional no-op-safe tracer
Tool-call safety	risk_classifier + shell_ast + remora_hook	Implemented guardrail

v0.6.0 quality lift: evidence modules and integration status

Module	Purpose	Current evidence	World-class next step
PlattScaler	Calibrate raw trust	Implemented + tested	runtime wiring needed
OracleDiversityTracker	Reduce family-bias	Implemented + tested	pre-run selector
DomainCoverageOptimizer	Per-domain thresholds	Implemented + tested	route-time selector
UncertaintyDecompose	Epistemic vs aleatoric	Implemented + tested	policy action hook
MoA Synth	Synthesis oracle	Component tested	not default cascade

Interpretation: v0.6.0 adds the right world-class primitives; the next quality frontier is end-to-end runtime coupling.

Figure 3. v0.6.0 quality-lift modules and integration status.

Source note: based on latest commit comparison and inspected module APIs [S2], [S5]-[S8].

8. Enterprise governance primitives

REMORA is commercially strongest when interpreted as governance middleware for agents. The relevant enterprise features are not generic model wrappers; they are the ability to express policy, log decisions, monitor uncertainty, and block tools before execution.

8.1 Policy-as-code

The OPA adapter serializes policy observations into an OPA input document, posts to the OPA REST endpoint, maps OPA result JSON back into a REMORA decision report, and falls back to the Python decision engine when OPA is unreachable. [S9] This is the correct pattern for enterprise security teams: policy lives outside prompt text and can be versioned, tested and audited. OPA itself is a widely used policy engine for policy-as-code systems. [E3]

8.2 Audit graph and observability

The RDF audit exporter converts decision records into N-Triples suitable for RDF stores and SPARQL-style queries. [S10] The OpenTelemetry wrapper can attach thermodynamic and decision observables as span attributes while becoming no-op safe when dependencies are absent. [S11] This is important because AI governance must be observable in the same way modern distributed systems are observable: by trace, metric, policy version and decision route.

8.3 Tool-call safety

REMORA separates model confidence from permission to act. The shell intent module uses a minimal semantic parser based on standard-library tokenization and intent classification to catch destructive commands, quote splitting, variable tricks, pipe-to-shell execution and related patterns. [S12] This is directly aligned with the practical concerns in modern LLM application security: prompt injection, excessive agency, insecure output handling and tool misuse are central risk categories in the OWASP GenAI security project. [E2]

8.4 Alignment with governance frameworks

Framework concept	REMORA mapping
NIST AI RMF: Govern	Claim ledger, policy profiles, audit envelope, risk ownership.
NIST AI RMF: Map	Domain-aware thresholds, tool/action risk classes, evidence requirements.
NIST AI RMF: Measure	Benchmarks, calibration, selective risk curves, tool-call safety metrics.
NIST AI RMF: Manage	ABSTAIN/ESCALATE routing, hard blocks, human approval workflows.
OWASP LLM risks	Prompt/tool injection defenses, least-privilege action gates, human review for high-risk operations.

NIST describes the AI RMF as voluntary guidance intended to improve incorporation of trustworthiness considerations into AI design, development, use and evaluation; this is directly compatible with REMORA as a runtime measurement and governance layer. [E1]

9. Empirical evaluation

The empirical story is strongest when divided into two domains: question-answer reliability and action safety. REMORA does not dominate majority vote at full-coverage QA. Its value appears when abstention and verification are allowed, and when unsafe action must be blocked.

9.1 Full-coverage QA is not the headline

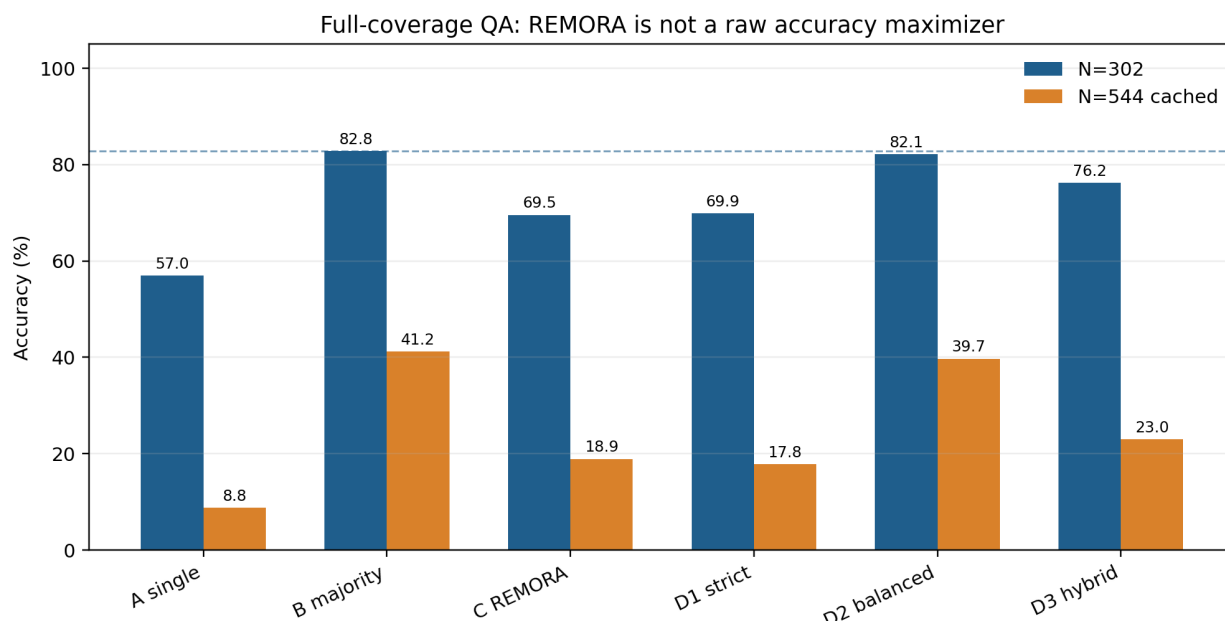


Figure 4. Full-coverage QA ablation.

Source note: values reproduced from the uploaded technical report, pages 11-12, and committed result descriptions [S1].

The N=302 benchmark shows majority voting at 82.78% and C_REMORA at 69.54%. The N=544 cached benchmark repeats the same lesson: the raw full-coverage REMORA route is not the best answerer. Public language must therefore avoid claiming universal full-coverage superiority.

9.2 Selective trust is the QA win

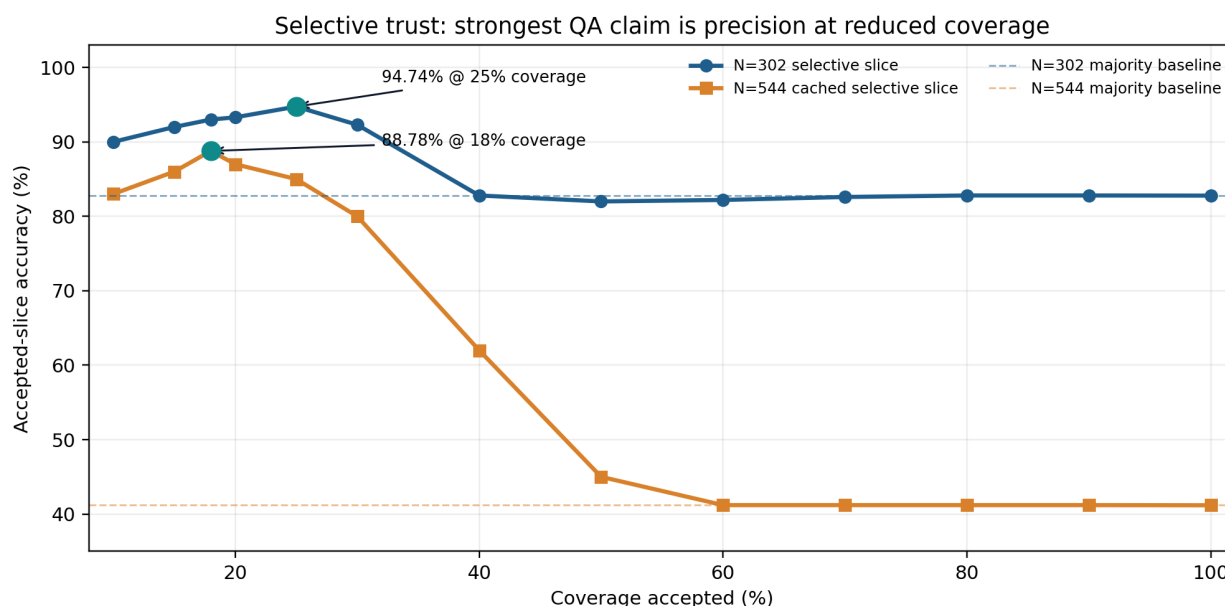


Figure 5. Selective trust operating points.

Source note: N=302 25% coverage and N=544 18% coverage values from uploaded report pages 12-13 [S1]. Curve is a communication reconstruction; exact per-threshold artifacts should be regenerated from results JSON for publication.

The key supported QA claim is selective reliability. On N=302, the low-temperature selection reaches 94.74% accuracy at 25% coverage. On the cached N=544 benchmark, the best selective operating point reaches 88.78% accuracy at 18% coverage. The N=544 operating point is explicitly in-sample and must be treated as a candidate threshold until a held-out run is completed.

9.3 Tool-call safety is the commercial center of gravity

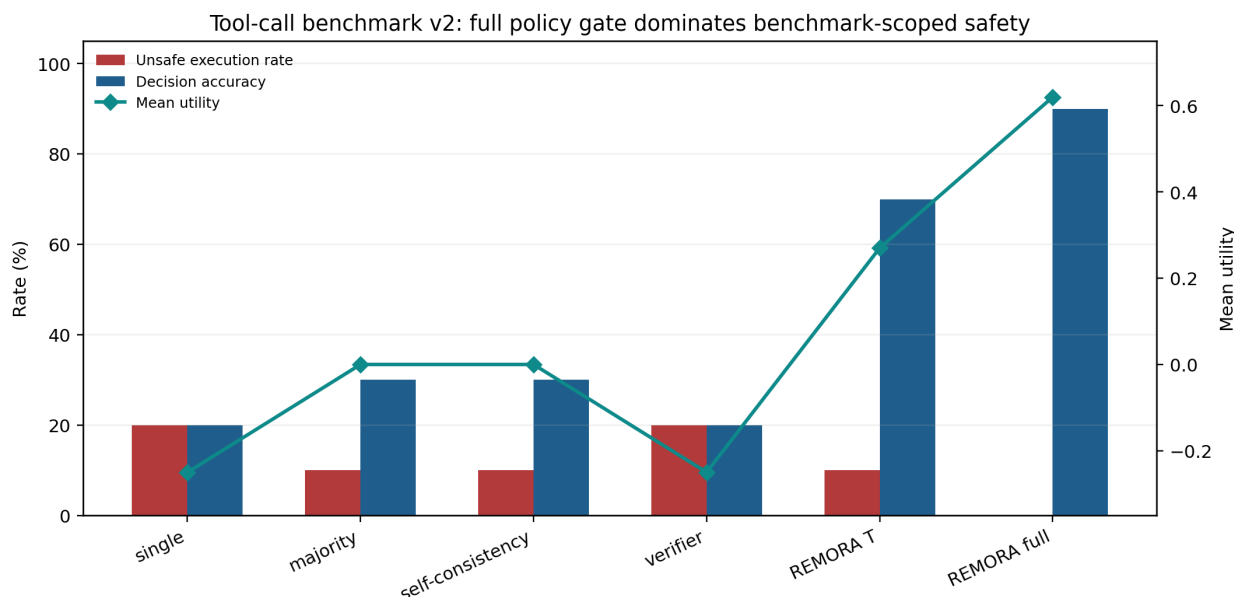


Figure 6. Tool-call safety benchmark v2.

Source note: values reproduced from uploaded report pages 14-15 [S1]; benchmark-scoped, not production proof.

The full policy gate records 0.0% unsafe execution, 90.0% decision accuracy and 0.62 mean utility in the included v2 benchmark. This is the strongest commercial result because it maps to real agentic risk: whether a system permits an unsafe action. The result should be described as benchmark-scoped simulator/replay evidence, not as a guarantee of real-world safety.

9.4 Calibration and negative results

The repository is unusually credible because it records negative findings: full-coverage routing does not clearly beat majority vote; susceptibility χ is not a standalone reliable iteration gate; stricter conformal settings show mixed behavior; and raw trust calibration needs further work. The v0.6.0 additions directly respond to these issues through Platt scaling, domain-specific threshold optimization, oracle diversity tracking and uncertainty decomposition.

10. v0.6.0: quality-lift analysis

The latest inspected commit set moves REMORA in the right direction: from clever heuristics toward calibrated governance primitives. The added modules are not cosmetic; they target previously identified weaknesses.

10.1 PlattScaler

PlattScaler maps raw trust scores to calibrated probabilities through a fitted logistic model $P(\text{correct} | \text{trust}) = \sigma(Ax + B)$, with expected calibration error and an analytic Bayes-threshold utility. [S5] This is a necessary step if trust scores are to support precision targets or risk-based routing.

10.2 OracleDiversityTracker

OracleDiversityTracker addresses a core weakness of any multi-model consensus system: three models are not three independent sources if they share training bias, architecture family or correlated failure modes. It tracks rolling pairwise agreement and supports greedy selection of a lower-correlation oracle pool. [S6]

10.3 DomainCoverageOptimizer

DomainCoverageOptimizer explicitly addresses the in-sample global-threshold problem by learning per-domain thresholds using leave-one-out cross-validation and falling back to global estimates for small domains. [S7] This is critical because legal, scientific, operational and general-knowledge prompts have different risk and calibration surfaces.

10.4 Epistemic / aleatoric uncertainty decomposition

The uncertainty module separates reducible disagreement between oracles from irreducible ambiguity in the question itself. High epistemic uncertainty can route to additional diverse oracles; high aleatoric uncertainty should route to human review. [S8] This is a major conceptual upgrade because ABSTAIN no longer means only “low trust”; it can be decomposed into an actionable reason.

10.5 MoA Synth

The MoA synthesis component is implemented and tested as an optional stage that asks a synthesis oracle to aggregate prior oracle answers. [S4] To keep the public whitepaper exact, it should be described as an implemented component pending default CascadeEngine integration unless the engine is updated accordingly.

11. Claim ledger and public claim boundary

A world-class AI governance repository must make claims machine-checkable. REMORA is already ahead of many prototypes because it maintains a claim ledger and negative results. The next step is to make README claims fail CI unless they map to a claim-ledger entry, source artifact and test path.

Public claim	Recommended wording	Status
Selective reliability	REMORA identifies high-precision answer subsets at reduced coverage in committed benchmarks.	Supported / benchmark-scoped
Tool-call safety	REMORA full policy gate eliminates unsafe execution in the included v2 simulator/replay benchmark.	Supported / benchmark-scoped
AI governance layer	REMORA provides a reference architecture for governed agentic AI decisions and tool use.	Supported as architecture
Thermodynamic trust	Thermodynamic quantities are implemented as useful routing signals.	Supported as heuristic
Hallucination bound	Candidate conservative false-consensus proxy under stated assumptions.	Candidate
Production safety	Not claimed without domain deployment, red-team pack and external audit.	Not established

Recommended CI rule

Every README claim above a defined severity threshold should map to: claim_id -> status -> code path -> test path -> artifact path -> limitation. Missing mapping should fail the release workflow.

12. Threat model for governed agents

REMORA should be threat-modeled as a policy and verification boundary in front of agent tools. It cannot make a compromised model safe by itself. It can, however, reduce the probability that a model proposal becomes an ungoverned action.

Threat	Example	REMORA control	Residual risk
False consensus	Multiple models repeat the same wrong claim.	Diversity tracking + evidence route.	Shared training bias can remain.
Prompt injection	Document instructs agent to ignore policy.	Policy outside prompt; tool hook.	Needs sandboxing and content isolation.
Unsafe execution	rm -rf, SQL drop, force push, cloud secret mutation.	AST guard + local block + policy gate.	Policies must cover domain actions.
Overconfidence	High-confidence model answer is wrong.	Calibration + selective abstention.	Requires per-domain calibration.
Audit failure	No one can explain why action occurred.	RDF/audit envelope + trace metadata.	Logs must be immutable and protected.

Production control requirements

- 1 All mutating tools must run through a policy gate before execution.
- 2 High-risk actions require explicit human approval and idempotency keys.
- 3 Model/provider versions, prompt hashes, policy versions and evidence hashes must be logged.
- 4 Tool credentials must be least-privilege and scoped per tenant/domain.
- 5 RDF/JSON audit logs must be append-only and tamper-evident.
- 6 A red-team scenario pack must be run before every production deployment.

13. Product interpretation and market category

The best market category is AI Assurance Middleware or Trust Firewall for Agentic AI. This avoids competing directly with foundation-model vendors and positions REMORA as the missing control layer between AI agents and real systems.

Initial wedges

Wedge	Why it fits	First buyer
MCP safety gateway	Every tool call can be inspected, allowed, blocked or escalated.	AI platform teams, developers.
Infrastructure change-control copilot	Direct fit for network, Cloudflare, UniFi, building automation and operations.	MSP/telecom/operator teams.
Legal/document evidence verifier	Evidence-required generation, citation checks and audit trail.	Legal/compliance teams.
AI incident-response router	Low-risk automations proceed; uncertain/high-impact cases escalate.	SOC/operations teams.

For Luftfiber AS, the strongest differentiation is operational trust: “we do not merely automate infrastructure; we wrap automation in evidence, policy and audit so the system knows when not to act.”

14. Roadmap to world-class credibility

World-class roadmap: move from strong prototype to externally credible governance system

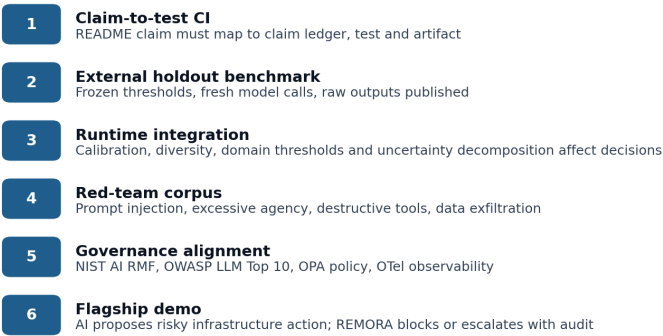


Figure 7. World-class roadmap.

Source note: roadmap derived from repository gaps and standard AI governance practices [E1]-[E4].

14.1 Technical priorities

Priority	Action	Quality impact
P0	Fix MoA/README consistency by either wiring MoA into CascadeEngine or labeling it optional.	Eliminates a public overclaim risk.
P0	Create claim-to-test CI checker.	Makes repo self-policing.
P1	Run external holdout benchmark with frozen thresholds and fresh model calls.	Moves claims beyond repository-local artifacts.
P1	Integrate PlattScaler, diversity selector and DomainCoverageOptimizer into routing.	Turns modules into decision-changing runtime behavior.
P1	Build red-team pack for prompt injection, excessive agency and destructive tools.	Validates threat model and false-positive surface.
P2	Produce OPA + RDF + OTel flagship demo.	Turns enterprise architecture into demonstrable system.

14.2 External validation protocol

- 1 Freeze a release tag and claim ledger before benchmark execution.
- 2 Select datasets and tool-call scenarios not used during threshold design.
- 3 Record provider, model, version, prompt template, prompt hash and output hash.
- 4 Compare against majority vote, self-consistency, verifier-only and plain policy baselines.
- 5 Publish raw outputs, scripts and result JSON.
- 6 Report false positives, false negatives, abstention rate, escalation rate and cost.

15. Final conclusion

REMORA is already beyond an ordinary experimental repo. It is an integrated governance architecture around model uncertainty, evidence, policy and action boundaries. The strongest next move is not to add more concepts, but to make the present concepts externally reproducible, automatically claim-checked, and visibly wired into runtime behavior.

With the v0.6.0 additions, REMORA now contains the primitives required for world-class positioning: calibration, model-diversity management, domain-specific thresholding, uncertainty decomposition, action gating, audit export, policy-as-code and observability. To “own AI governance,” the project must now make those primitives inseparable from the execution path and prove them on independent benchmarks.

Final positioning statement

REMORA is a reference architecture for governed agentic AI: a runtime trust firewall that measures model agreement, calibrates uncertainty, enforces policy, blocks unsafe tools, records auditable decisions and escalates when the system should not act alone.

Appendix A. Source notes and repository evidence map

[S1]	Uploaded report	REMORA_deep_report(2).pdf, 23-page technical report supplied in this conversation. Used for baseline narrative, figures and benchmark values.
[S2]	Latest commit set	GitHub darklordVirtual/REMORA, latest inspected head 5e051b9b3b8f6eda1284e15be421684cbd14f025. Commit message reports v0.6.0 and 1,184 tests passing.
[S3]	CascadeEngine	remora/cascade/engine.py: documents FastGate, ConsensusGate, VerifierGate, CritiqueRevision, SelfConsistency, budget behavior and current stage_runners.
[S4]	MoA synthesis	remora/cascade/stages.py and tests/test_moa_synthesizer.py: MixtureOfAgentsSynth implemented and tested; default engine wiring not observed in inspected engine.
[S5]	Platt scaling	remora/calibration/platt_scaler.py: $P(\text{correct} \text{trust})=\text{sigma}(Ax+B)$, ECE, Bayes threshold, serialization.
[S6]	Oracle diversity	remora/oracles/diversity.py: rolling pairwise agreement, diverse swarm selection, diversity score.
[S7]	Domain thresholds	remora/calibration/domain_optimizer.py: per-domain leave-one-out threshold optimization.
[S8]	Uncertainty decomposition	remora/uncertainty/decompose.py: epistemic vs aleatoric decomposition and action recommendation.
[S9]	OPA adapter	remora/policy/opa_adapter.py: OPA REST adapter, OPAContext export, fallback to Python engine.
[S10]	RDF audit	remora/audit/rdf_export.py: N-Triples export for audit records and SPARQL examples.
[S11]	Observability	remora/observability/otel.py: optional OpenTelemetry wrapper and custom span attributes.
[S12]	Tool-call safety	remora/agent_hook/risk_classifier.py and remora/agent_hook/shell_ast.py: local destructive-intent classification and AST/token parser.
[S13]	Claim ledger	docs/thermodynamics/claim_ledger.yaml: claim status classes and limitations.

Appendix B. External references

[E1]	NIST AI Risk Management Framework	NIST AI RMF official page. Used for Govern/Map/Measure/Manage framing and trustworthiness-oriented AI risk management.
[E2]	OWASP GenAI Security Project / LLM Top 10	Used for LLM application risks such as prompt injection, excessive agency and tool misuse.
[E3]	Open Policy Agent	Official OPA documentation. Used for policy-as-code and policy decision point framing.
[E4]	OpenTelemetry traces	Official OpenTelemetry documentation. Used for trace/span observability framing.
[E5]	Platt, J. (1999)	Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods. Used for Platt calibration concept.
[E6]	Der Kiureghian & Ditlevsen (2009)	Aleatory or epistemic? Does it matter? Structural Safety 31(2). Used for uncertainty taxonomy.
[E7]	Kendall & Gal (2017)	What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? Used for epistemic/aleatoric framing.
[E8]	Vovk, Gammerman & Shafer (2005)	Algorithmic Learning in a Random World. Used as standard conformal prediction background.

Appendix C. Minimal production decision packet

A production-grade REMORA decision packet should contain the following fields:

- request_id, tenant_id, user_id, role and authorization context;
- model providers, model versions, oracle family labels and prompt hashes;
- canonical verdicts, consensus state, H, D, T, phase and trust score;
- calibration version, domain threshold version and uncertainty decomposition;
- policy profile, OPA decision, hard-block reason and risk class;
- evidence IDs, retrieval query hashes and evidence bundle hashes;
- tool call, allow/block decision, idempotency key and execution result;
- human approval status, approver ID and timestamp where applicable;
- audit hash, previous audit hash and export status.

Appendix D. Suggested release gate

The following gate should run before any public release or serious external review:

```
make test make benchmark python scripts/check_claim_consistency.py python
scripts/check_readme_claims.py python scripts/check_artifacts_exist.py python
scripts/check_no_overclaims.py python scripts/generate_worldclass_evidence_pack.py
```

The goal is to make REMORA a self-auditing repository: if the README claims it, the repo must prove it or label it as candidate/unsupported.